PRIME NUMBERS

# The Prime Number Theorem

Primes become rarer among larger integers, and the theorem says precisely how.

THEOREM

With $\pi(n)$ the number of primes up to $n$: $\pi(n) \approx \dfrac{n}{\ln n}$ as $n$ grows large.

1 **COUNT**

Among the first $100$ integers, $25$ are prime; among the first $1{,}000$, $168$; among the first $1{,}000{,}000$, $78{,}498$.

2 **APPROXIMATE**

$\pi(n) \approx n/\ln n$.

3 **RELATIVE ACCURACY**

The ratio $\dfrac{\pi(n)}{n/\ln n}$ approaches $1$.

4 **AVERAGE GAP**

Near $n$, roughly one in every $\ln n$ integers is prime: about $13.8$ near $10^6$, about $27.6$ near $10^{12}$.